

• , f „ „ † ‡ ^ ‡ ‡

%Š-4 ‡<Œ • ž š • Œ

(Mini Processor Datapath using Logisim Evolution)

• ‘ 1447 / ’ 2026	:	’
“ ‡ ” ’ f • -	:—” „ • ~	
• ™ ‡ š / š ›	:œ ‡ • ~ ‰	

† ‡ ^ ‡ ž .1

. € • , f „ ... † ‡ ^ † (Sequential Circuits) ħ ,š ‡ < ,
ĈĈ... ALU • † (Combinational Circuits) ž† , ,š • ħ • ‘ „ ’ “ .Logisim Evolution ” ,ħ
• • ħ –ħ-4 — , ħ ħ (Mini Processor Datapath) ,™ š ” ħ š > ħ œ ^ € • ‘
š • ‘ ... ž žŸ ħ Ć ,£ „ ’ ¢ ¥

> ’ Ÿ ž .2

. (AND, OR, XOR, ADD, SUB) • Ÿ ħ „ ... : (ALU) žŸ ħ š “
. (Clock) ©ž • <• Result Register ‡ ^ € • <• Register B Register A • ^ Ć
: (Registers) ^ “
. ħ š Ć « ^ Ć š ħ Decoder 3:8 • • ħ (Opcode) • † < , - ... ħ : (Control Unit) Ć
ħ š “
. ħ ® š ... ħ š ħ Carry Flag ,™ ħ š ħ Zero Flag £ : (Flags) š ħ “

> ‡ € ” ħ .3

(Full Adder) Ć £ ‡ „ 3.1

: ’ Ć “ ħ ° • Ÿ ± •

$$\text{Sum} = A \oplus B \oplus \text{Cin}$$

$$\text{Cout} = (A \cdot B) + (\text{Cin} \cdot (A \oplus B))$$

Cout	Sum	Cin	B	A
0	0	0	0	0
0	1	0	0	1
0	1	0	1	0
1	0	0	1	1
0	1	1	0	0
1	0	1	0	1
1	0	1	1	0
1	1	1	1	1

Opcode „žŹ € .4

: ^ ALU • 2 3 • € š ¬i i (Opcode) !,¬¬ ! (Decoder 3:8) < , - €... •

ž•	Sub žž” •~	RegWrite ¢ • §	ALU_Sel ”	(Operation)	Opcode ”
¬! ! ¬! ž AND	0	1	000	AND	000
¬! ! ¬! ž OR	0	1	001	OR	001
(’ •) ž XOR	0	1	010	XOR	010
(A + B) ¬!-4 § ’ ! § “ 3	0	1	011	ADD	011
(A - B) § • • ! ! § μ,¶	1	1	100	SUB	100
(NOP) • • , .	0	0	101	RESERVED	101
(NOP) • • , .	0	0	110	RESERVED	110
(NOP) • • , .	0	0	111	RESERVED	111

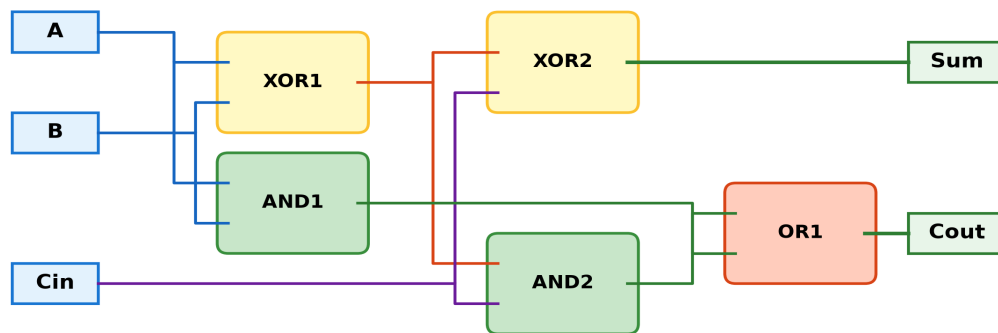
Logisim Evolution €

””! .5

(Full Adder Subcircuit) ¢

£ € §”! 5.1

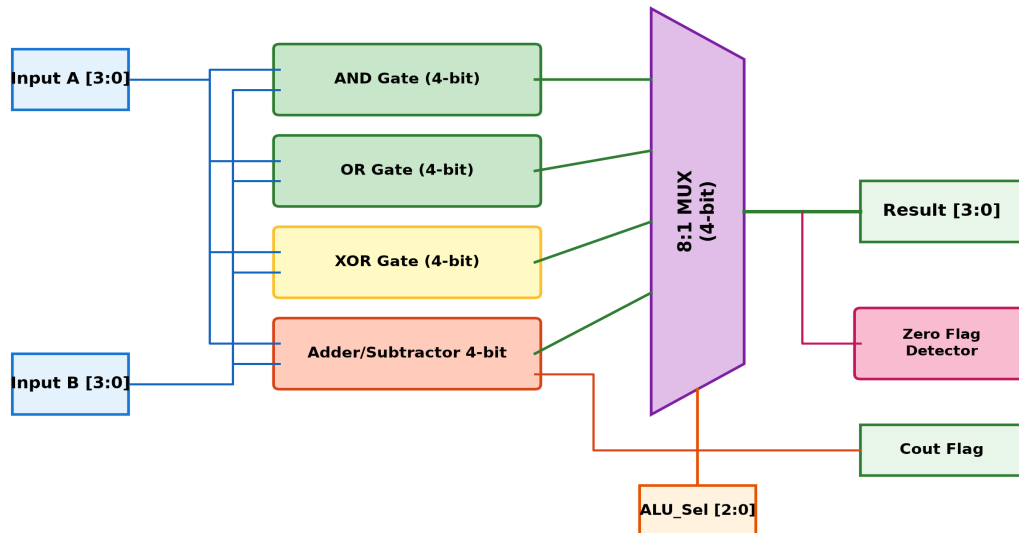
Full Adder Circuit Schema (دائرة الجمع الكامل)



(ALU 4-Bit) ” , • f € 5.2

:ALU_Sel , Ÿ€ > | | Ÿ Multiplexer 8:1 § • . , , -4 —, | • €
ALU • ž

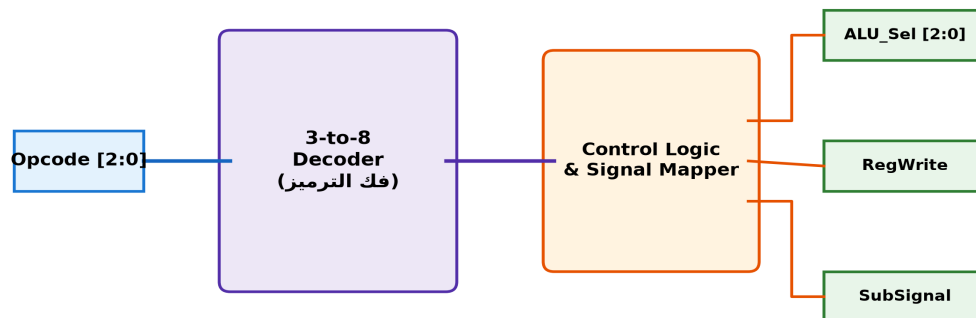
4-Bit ALU Internal Architecture (وحدة الحساب والمنطق)



(Control Unit & Decoder) © ‡ a - f € 5.3

:µ, Ÿ SubSignal |š ¬j • <• RegWrite |š ¬j Ÿ | | ¥ ‡|• , , Ÿ€ 8 | Opcode | , ¬ œ 1 ,
Decoder 3:8 € | § 1

Control Unit Schematics (وحدة التحكم والفك)

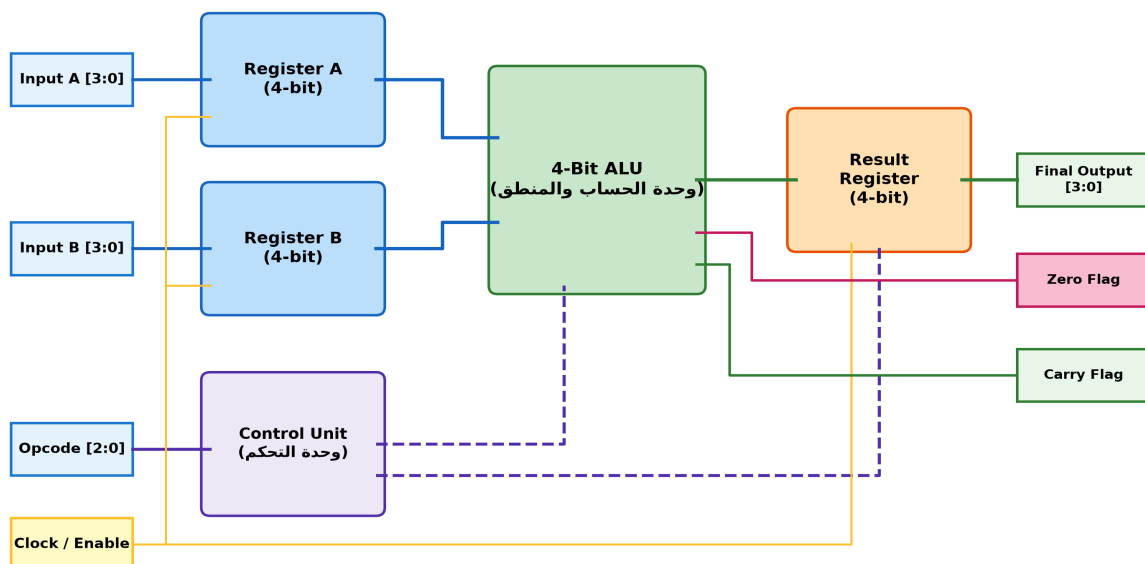


(Full Mini Processor Datapath) ¢ ^ ' ¨ §”! 5.4

:Clock ©Ž ¨ ¨ • < “ Result Register

Control Unit € | § ALU | § B A
^ • ¨ € ¨ , ¨ ¨ • »¼

Mini Processor Datapath System (المخطط الشامل للمشروع)



ž « - ¢ <^ ”¬ ¥‡• .6

.B A ^ ½ ¥ <• ©Ž “ Load_B Load_A ... Input B Input A ½ œ €•j : ž

.1

.RegWrite = 1 ALU_Sel = 011 š¬j € | § ½ ‡Opcode € ½ (ADD 011 ^) • † €
:, • |,¬ - ½ .2

.Result Register € j Multiplexer ,ž ’, , š ½ ¨ ALU • • :ALU • ¨ „3

.Final Output ’š ¥¼j Result Register ½ š ¥ • <• ¨ RegWrite |š¬j • ‡ ©Ž ½ § “ :

• <• .4

- ® ! .7

. ž š ž ¥... «< ‡ ž • ™ |> ...† ALU • “
šž €± - ž. Logisim Evolution À ¨ μ ¨ € | § ^ ALU • ¨ “ € ,™ š • ‘

> ¨ 2 ½ Á ¨